Heuristic Comparison: SLD vs Manhattan Distance (A*)

Exam-style numerical question + complete worked solution

Problem Setup

Goal: Bucharest B at coordinate $(0, 0)$.

Note: The map/graph is assumed given in the course material. Use the coordinates and edge costs below for the numerical checks.

Coordinates

Node	Coordinates (x, y)
A	$(-285, 230)$
S	$(-200, 155)$
R	$(-168, 95)$
P	$(-70, 69)$
F	$(-160, 78)$
C	$(-120, -106)$
T	$(-300, 135)$
Z	$(-300, 223)$
O	$(-320, 205)$
B	$(0, 0)$

Selected Edge Costs

- $R \to P = 97$
- $S \to R = 80$
- $F \to B = 211$
- $A \to S = 140$

Heuristics

- **SLD heuristic** (given table; assume admissible and consistent).
- **Manhattan heuristic:**

$$h_{\text{MAN}}(n) = |x_n| + |y_n|$$

because the goal is at $(0, 0)$.

Questions

Q1. Compute Manhattan heuristic values

Compute $h_{\text{MAN}}(n)$ for:

1. R

2. P
3. F

Q2. Check consistency of Manhattan heuristic

A heuristic is consistent if for every edge (n, n'):

$$h(n) \leq c(n, n') + h(n').$$

Check consistency of h_{MAN} on:

1. Edge $R \to P$
2. Edge $S \to R$
3. Edge $F \to B$

State clearly whether each inequality holds.

Q3. Check admissibility of Manhattan heuristic

A heuristic is admissible if:

$$h(n) \leq h^*(n),$$

where $h^*(n)$ is the true shortest path cost to the goal.

1. Compute the true shortest path cost from F to B.
2. Compare with $h_{\text{MAN}}(F)$.
3. Is h_{MAN} admissible? Justify numerically.

Q4. Dominance Question

Recall: Heuristic h_2 dominates h_1 if $h_2(n) \geq h_1(n)$ for all n, and both are admissible.

1. Compare $h_{\text{MAN}}(F)$ and $h_{\text{SLD}}(F)$.
2. Can we conclude that Manhattan dominates SLD for A*? Justify carefully using admissibility.

Complete Worked Solution

Solution to Q1

Since $B = (0, 0)$, we use $h_{\text{MAN}}(n) = |x| + |y|$.

- $R(-168, 95)$:
$$h_{\text{MAN}}(R) = |-168| + |95| = 168 + 95 = 263.$$
- $P(-70, 69)$:
$$h_{\text{MAN}}(P) = |-70| + |69| = 70 + 69 = 139.$$
- $F(-160, 78)$:
$$h_{\text{MAN}}(F) = |-160| + |78| = 160 + 78 = 238.$$

Solution to Q2 (Consistency)

Consistency condition:

$$h(n) \leq c(n, n') + h(n').$$

(a) Edge $R \to P$ (cost 97).

$$h(R) = 263, \qquad c(R, P) + h(P) = 97 + 139 = 236.$$

Check:

$$263 \leq 236 \quad \text{is false.}$$

Thus h_{MAN} is **inconsistent** on $R \to P$.

(b) Edge $S \to R$ (cost 80). First compute:

$$h(S) = |-200| + |155| = 200 + 155 = 355.$$

Then:

$$c(S, R) + h(R) = 80 + 263 = 343.$$

Check:

$$355 \leq 343 \quad \text{is false.}$$

Thus h_{MAN} is **inconsistent** on $S \to R$.

(c) Edge $F \to B$ (cost 211).

$$h(F) = 238, \qquad h(B) = 0, \qquad c(F, B) + h(B) = 211 + 0 = 211.$$

Check:

$$238 \leq 211 \quad \text{is false.}$$

Thus h_{MAN} is **inconsistent** on $F \to B$.

Conclusion: h_{MAN} is not consistent on this weighted road graph.

Solution to Q3 (Admissibility)

Admissibility requires $h(n) \leq h^*(n)$.

- The graph includes a direct edge $F \to B$ with cost 211.
- Therefore the true shortest cost is $h^*(F) = 211$.
- But $h_{\text{MAN}}(F) = 238$.

Since

$$238 > 211,$$

Manhattan overestimates the true cost.

Conclusion: h_{MAN} is **not admissible** (counterexample at F).

Solution to Q4 (Dominance)

From the SLD table, $h_{\text{SLD}}(F) = 178$. We computed $h_{\text{MAN}}(F) = 238$, hence

$$h_{\text{MAN}}(F) > h_{\text{SLD}}(F).$$

However, the dominance definition used with A* requires *both* heuristics to be admissible. Since h_{MAN} is **not admissible**, we **cannot** conclude it dominates SLD in the A* sense.

Final conclusion: Even if Manhattan is larger numerically, it does not dominate SLD for A* because it fails admissibility (and also fails consistency).